

detect intrusion or security breaches without manual intervention.

The challenges faced

Electro-optics require a good knowledge of electronics, mechanical engineering, optics and programming. To get engineers with strong mechanical and electronics design know-how was a challenge for Tonbo, given the high number of software engineering candidates at the time.

Finding people familiar with optics was more challenging. Kumar says, "There are barely any schools or colleges in India that have specialised focus on optics at a fundamental level. We were heavily dependent on freshers and had to train them to the level where they could start contributing technically."

Hardcore trainings were carried out on-the-job, in the field. "We encouraged freshers to be available on field during interactions with clients," Kumar adds.

Lack of a strong hardware ecosystem also factored in the development of the products. Evaluating real-time on-field conditions was also not easy.

Kumar recollects, "The initial few years were spent on building a lot of technology to generate the IP. Products started coming out in 2012. The next few years went into taking these to customers and deploying these."

Tonbo Imaging is venturing out into new domains. Kumar mentions, "We are now supplying thermal imaging for critical installations, like nuclear plants, airports, oil fields in addition to smart cities and medical electronics (for example, for breast cancer detection). We are also talking to Indian automotive manufacturers and Tier-I suppliers to integrate thermal imaging into non-premium Indian vehicles."

The company also plans to create platforms for devices where their electro-optics can be used. With offices in Bengaluru and Florida (USA) and one set to open in Lithuania, it is starting to mature from being a startup. However, it plans to move ahead with the same zeal, as Kumar believes, "The key advantage of being a startup is being agile and not being slowed down with processes."

—Paromik Chakraborty, technical journalist, EYF

PlutoX: DIY Drones Made Flexible And Simple

Electronics enthusiasts and hobbyists love to fiddle with technology, and a lot of the time end up creating something useful for real life. Of the many popular electronic projects out there, UAV kits, like drones, draw special interest. Drones can be used in a variety of use cases and, in turn, generate a great business proposition for builders. However, UAV kits available in the market today often come with many complexities for the DIYer community. Mumbai-based Drona Aviation, founded by IIT Mumbai alumni, Dinesh Sain, Prasanna Shevare and Apurva Godbole, took up this challenge and turned it into a business of their own.

Shevare and Sain have been working with drones for a long time. While participating in drone competitions or other design activities, they identified that it took extremely long to build simple UAV applications. "Drones available are fragile and difficult to program, which cause many challenges for developers," says Godbole.

In 2014, Drona Aviation was founded. Exploring the various challenges surrounding the technology, the team created its own UAV platform, called Pluto, which was fit for educational purposes, workshops and basic uses.

However, demands rose, and users wanted programmable drones with more features. Godbole recalls, "Customers wanted the option to be able to add more hardware on top of existing set ups."

With that perspective, the team came up with an upgraded kit, called PlutoX. The first prototype was launched in February 2018. Tinkerers and developers from India and abroad were handed prototypes for trial. These are now available for pre-order commercially.

The complete PlutoX box consists of a ready-to-fly nano-drone, spare propellers, an image processing camera, range sensors, DC motors, a breakout board and an app controller. The core electronics board, called PrimusX, has been designed and manufactured by the Drona team.

To accommodate extra hardware, a unibus has been added to the board. It has 16 general-purpose input-output (GPIO) pins in one place. Using this port, users can add external hardware (such as breakout board) provided in PlutoX box to the unit, or use their own external hardware in a simple manner.

The camera can record photos and videos in HD quality—files can



PlutoX box contents